

CHP-5 Capacitors and Dielectrics

General Physics 2

2026-09-30

Capacitance

Capacitors and Dielectrics

An arrangement of conductors used to store electric charge is called a **capacitor**.

First, we will study capacitors and their properties.

Then we will study electric fields in regions filled with an insulating material, or **dielectric**.

Capacitance of an Isolated Sphere

Recall the potential of an isolated metallic sphere of radius R carrying charge Q , measured relative to infinity:

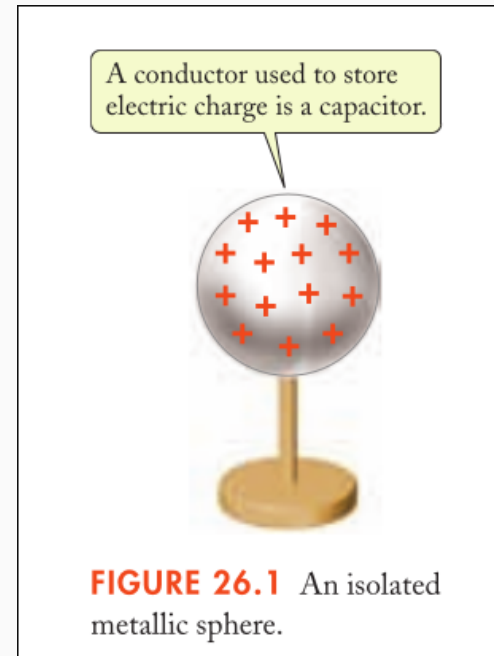
$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{R}$$

We can write this relation as

$$Q = CV$$

The constant C is the **capacitance**. For the sphere,

$$C = \frac{Q}{V} = 4\pi\epsilon_0 R$$



Units of Capacitance

The SI unit of capacitance is the farad:

$$1 \text{ F} = 1 \frac{\text{C}}{\text{V}}$$

A farad is a large unit, so common practical units include

$$1 \text{ }\mu\text{F} = 10^{-6} \text{ F}, \quad 1 \text{ pF} = 10^{-12} \text{ F}.$$

The vacuum permittivity can also be expressed in farads per meter:

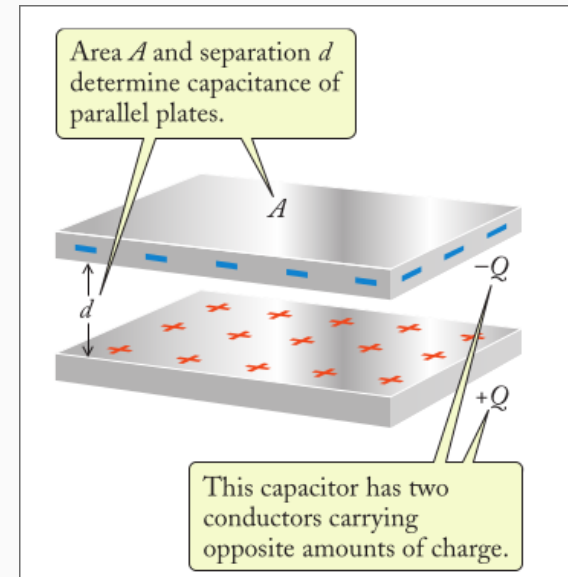
$$\epsilon_0 \approx 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N} \cdot \text{m}^2} = 8.85 \times 10^{-12} \text{ F/m}$$

A Capacitor with Two Conductors

A common capacitor consists of two conductors insulated from each other and carrying equal and opposite charges, $+Q$ and $-Q$.

$$Q = C\Delta V$$

Here Q is the **magnitude** of the charge on either conductor, and $\Delta V = V_+ - V_- > 0$.



The Parallel-Plate Capacitor

Two large parallel plates have area A , separation d , and charges $+Q$ and $-Q$. Neglect edge effects.

The field between the plates is

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}$$

Integrating from the negative plate to the positive plate gives

$$\Delta V = V_+ - V_- = - \int_-^+ \vec{E} \cdot d\vec{l} = Ed = \frac{Qd}{\epsilon_0 A}$$

$$C = \frac{Q}{\Delta V} = \frac{\epsilon_0 A}{d}$$

A larger plate area and a smaller separation give a larger capacitance.

The Cylindrical Capacitor

Consider two long coaxial conductors of length L : an inner cylinder of radius a carrying $+Q$, and an outer cylindrical shell of inner radius b carrying $-Q$, with $a < b$. Neglect end effects.

Choose a coaxial cylindrical Gaussian surface with $a < r < b$:

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

$$E(2\pi rL) = \frac{Q}{\epsilon_0}$$

$$E(r) = \frac{Q}{2\pi\epsilon_0 L r}$$

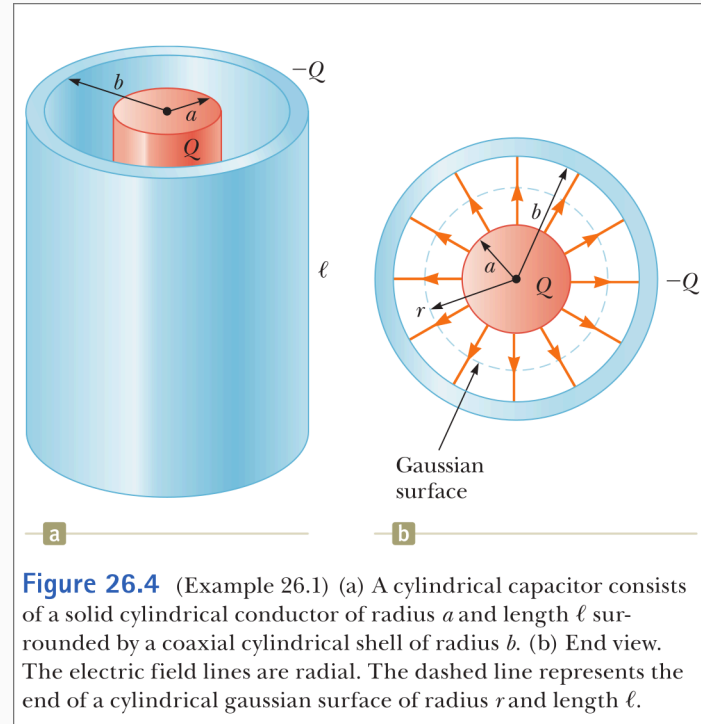


Figure 26.4 (Example 26.1) (a) A cylindrical capacitor consists of a solid cylindrical conductor of radius a and length ℓ surrounded by a coaxial cylindrical shell of radius b . (b) End view. The electric field lines are radial. The dashed line represents the end of a cylindrical gaussian surface of radius r and length ℓ .

Capacitance of Coaxial Cylinders

Compute the positive potential difference from the outer cylinder to the inner:

$$\begin{aligned}\Delta V &= V(a) - V(b) = - \int_b^a E(r) dr \\ &= - \frac{Q}{2\pi\epsilon_0 L} \int_b^a \frac{dr}{r} \\ &= \frac{Q}{2\pi\epsilon_0 L} \ln\left(\frac{b}{a}\right)\end{aligned}$$

Hence,

$$C = \frac{Q}{\Delta V} = \frac{2\pi\epsilon_0 L}{\ln\left(\frac{b}{a}\right)}$$

Capacitors in Combination

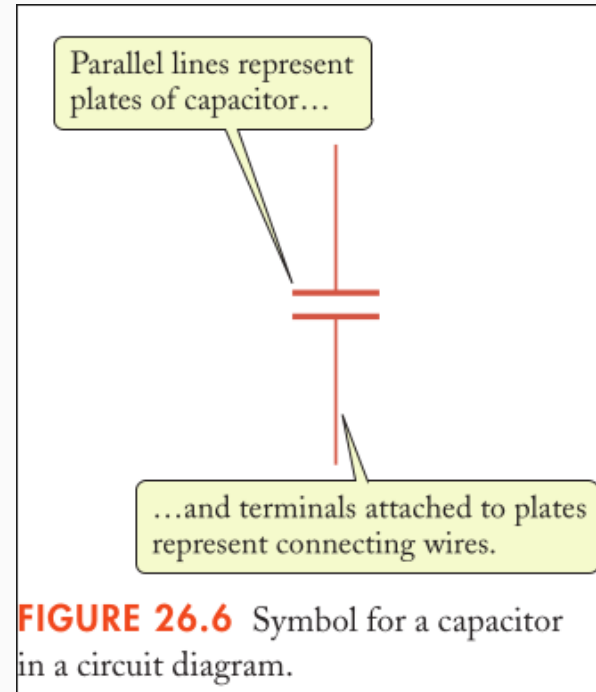
Capacitors in Circuits

Capacitors are used in practical circuits, including radios and amplifiers. Their circuit symbol consists of two separated parallel lines, with one connection to each plate.

Capacitors may be connected in **parallel** or in **series**.

- In parallel, each capacitor connects across the same two nodes.
- In series, capacitors connect end to end along a single branch.

We replace a combination by an **equivalent capacitance** C satisfying $Q = C\Delta V$ at the external terminals.



Parallel Combination

Corresponding plates of C_1 and C_2 are joined by wires, so the potential difference across both capacitors is the same:

$$\Delta V = \frac{Q_1}{C_1} = \frac{Q_2}{C_2}$$

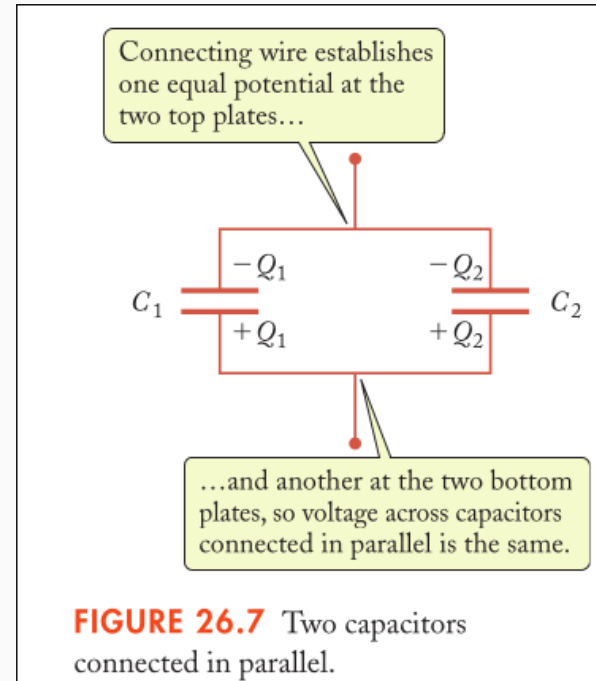
The charge supplied to the positive terminal is

$$Q = Q_1 + Q_2 = C_1\Delta V + C_2\Delta V = (C_1 + C_2)\Delta V$$

$$C = C_1 + C_2$$

For any number of capacitors in parallel,

$$C = C_1 + C_2 + C_3 + \dots$$



Series Combination

Connect C_1 and C_2 end to end then both capacitors acquire the same charge magnitude Q .

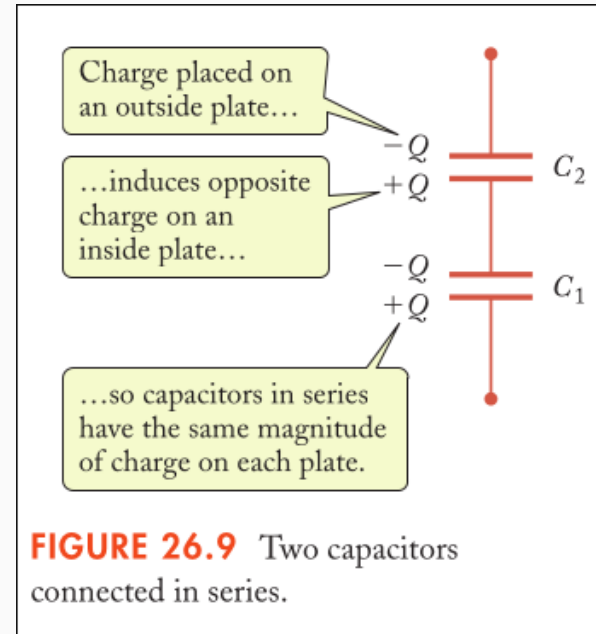
$$\Delta V_1 = \frac{Q}{C_1}, \quad \Delta V_2 = \frac{Q}{C_2}$$

The total potential drop is

$$\Delta V = \Delta V_1 + \Delta V_2 = Q \left(\frac{1}{C_1} + \frac{1}{C_2} \right) = \frac{Q}{C}$$

For any number of capacitors in series,

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$



Example: A Mixed Combination

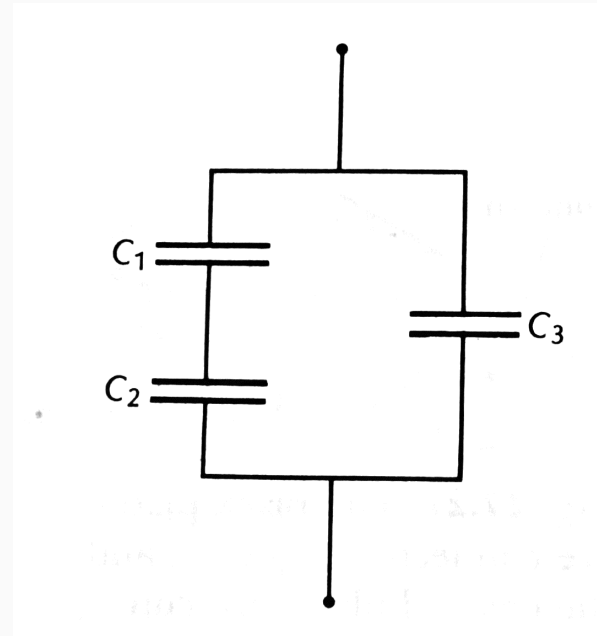
Two capacitors C_1 and C_2 are in series in one branch. That branch is connected in parallel with a third capacitor C_3 .

First replace the series pair:

$$C_{12} = \left(\frac{1}{C_1} + \frac{1}{C_2} \right)^{-1} = \frac{C_1 C_2}{C_1 + C_2}$$

Then add the parallel branch:

$$C = C_3 + \frac{C_1 C_2}{C_1 + C_2}$$



Example: Reconnecting Charged Capacitors

Two capacitors, $C_1 = 2.0 \mu\text{F}$ and $C_2 = 6.0 \mu\text{F}$, are each charged by a 24 V battery. The battery is then removed. The positive plate of each capacitor is connected to the negative plate of the other, making a closed loop with two shared nodes. What is the final charge on each capacitor after redistribution?

The initial charge magnitudes are

$$Q_1 = C_1 V_0 = 48 \mu\text{C}, \quad Q_2 = C_2 V_0 = 144 \mu\text{C}.$$

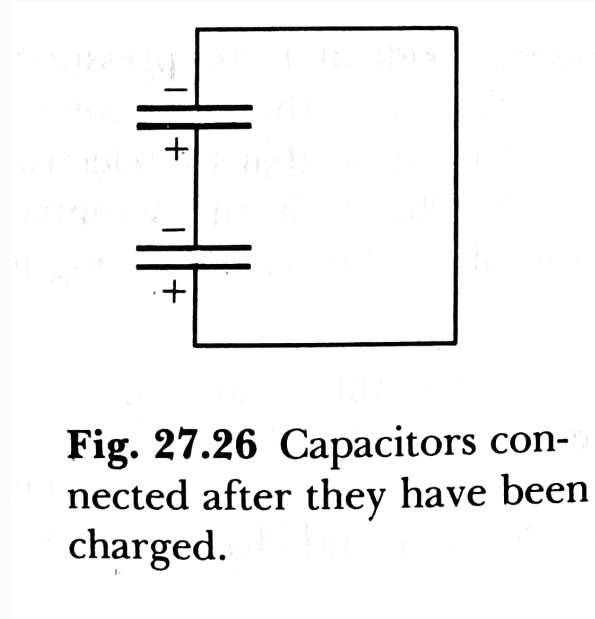


Fig. 27.26 Capacitors connected after they have been charged.

Conserve Charge on Each Connected Node

The node joining the initially negative plate of C_1 to the initially positive plate of C_2 has total charge

$$Q_2 - Q_1 = (C_2 - C_1)V_0.$$

Let the final positive plate charges on this node be q_1 and q_2 . Then

$$q_1 + q_2 = Q_2 - Q_1.$$

Both capacitors now connect across the same two nodes, so their final voltage magnitudes are equal:

$$\frac{q_1}{C_1} = \frac{q_2}{C_2} = V_f$$

The smaller capacitor reverses polarity in this example.

Final Charges After Redistribution

Combining charge conservation and the common final voltage,

$$V_f = \frac{(C_2 - C_1)V_0}{C_1 + C_2} = 12 \text{ V.}$$

Equivalently,

$$q_1 = \frac{V_0(C_2 - C_1)}{1 + \frac{C_2}{C_1}}, \quad q_2 = \frac{V_0(C_2 - C_1)}{1 + \frac{C_1}{C_2}}.$$

Substituting the values,

$$q_1 = 2.4 \times 10^{-5} \text{ C}, \quad q_2 = 7.2 \times 10^{-5} \text{ C}$$

These are charge magnitudes; the opposite plate of each capacitor carries the corresponding negative charge.